IKPM Portal

Final Change Report

49 numbered changes in 4 phases and 14 steps. 46 are to be built, because 3 items from your gap list are replaced by better versions (34, 35, 36 by 42, 43, 44). Built from your gap list, your SRS and the code in IKPM-Portal.zip.

Phase	Name	Change numbers	Numbered	To build
Phase 0	Foundation. Build first.	46, 47	2	2
Phase 1	Fixes to existing modules	1 to 28, 39, 40, 41, 48, 49	33	33
Phase 2	New features for existing modules	29 to 32, 45	5	5
Phase 3	New modules	33 to 38, 42, 43, 44	9	6
Total			**49**	**46**

Please send me 9 items I could not read. Page 1 of your gap list PDF is blank, so items 1 to 7 are missing (item 7 is readable only from its last lines). The PDF also ends after item 36, but your Phase 3 header says 6 changes, which means items 37 and 38 exist. That is why my new changes start at 39. Send items 1 to 7, 37 and 38 as text or a screenshot and I will place them in the right steps. They are counted in the totals above but marked "not readable" below.

NEW = added by this report (detail pages at the end). REPLACED = your item, superseded by a new item. Phase 1 count assumes your Phase 1 is items 1 to 28.

PHASE 0 — Foundation: roles, security, IDs — 2 changes

Step 0.1 Roles, modules, routes and security rules

No.	Change	Where
46	**NEW** Add modules (hr, attendance, payroll, challans, search), new Accounts / Finance and HR roles, routes, Firestore rules, indexes, tests	`permissions.ts, firestore.rules, dataService.ts, App.tsx`

Step 0.2 Safe ID counters

No.	Change	Where
47	**NEW** Counter document in a Firestore transaction instead of "list length + 1"; generateDocumentId	`dataService.ts, formatters.ts`

PHASE 1 — Fixes to existing modules — 33 changes

Step 1.1 Lead pipeline and calls (CRM)

No.	Change	Where
1 to 7	Not readable in the uploaded PDF (page 1 blank). Item 7 ends with: replace PipelineStage type and PIPELINE array with the SRS stage list	`leadWorkflow.ts`
8	inspection_complete can be set only by Operation / Operation Coordinator. CRO must not	`leadWorkflow.ts, lead detail page`
9	Lock "Create Proposal" until the lead is inspection_complete	`proposal action`
10	Add already_has_vendor outcome; every call log must have an outcome	`leadWorkflow.ts OUTCOMES, call log form`

Step 1.2 Operations: PCO records, coordinator reports, challan, filing

Do in this order. Item 48 needs the basic employee record from item 42 (name, phone, role, status), so build that small part of 42 first. Gap items 11 to 14 are now steps the coordinator completes for the PCO (item 49).

No.	Change	Where
48	**NEW** PCOs are enrolled records with no login; jobs are assigned to them. Fix the double-booking check	`JobsListPage.tsx, dataService.ts, operations.ts`
49	**NEW** Coordinator submits the full report for the PCO; the submitter cannot verify the same report	`JobsListPage.tsx, pdfGenerator.ts`
11	Pre-service checklist barrier: uniform, safety gear, chemicals issued. All ticked	`JobsListPage.tsx`
12	At least 1 before photo before the job can start	`JobsListPage.tsx`
13	Signed service report photo required before submit	`JobsListPage.tsx`
14	Late submission flag and lateDays, visible in verification	`job record, verification view`
15	Chemical match: PCO-reported chemicals shown next to what the coordinator enters	`JobsListPage.tsx verifyForm`
16	Risk Assessment form after verification; B2B requirement notifies CRO	`JobsListPage.tsx, notifications`
17	Job status flow: assigned, pre_check_done, in_progress, submitted, verified or issue, closed	`workflow.ts`
39	**NEW** Challan number and challan file required to confirm (verify) a report	`JobsListPage.tsx, operations.ts`
40	**NEW** Challan register, searchable by number	`ChallansPage.tsx (new)`

No.	Change	Where
41	**NEW** Confirmed report, signed report photo and challan saved in the client and service folder; failed filing is shown and retried	`JobsListPage.tsx, folders.ts`

Step 1.3 Complaints

No.	Change	Where
18	Auto-assign new complaints to the Operations Manager role	`ComplaintsPage.tsx`
19	Notify the CEO on every new complaint	`ComplaintsPage.tsx`
20	Wire overdue escalation to the CEO (isComplaintOverdue)	`complaints.ts, notifications`
21	Closing note required to close	`ComplaintsPage.tsx`
22	Add escalated status; fix categories (re_infestation, pco_behaviour, service_quality)	`complaints.ts`

Step 1.4 Finance

No.	Change	Where
23	FBR billing profiles in Settings (super admin only): Sales Tax 15% WHT, Cash 5.5% WHT; replace provincial rates	`SettingsPage.tsx, finance.ts`
24	Auto-select the FBR profile from the client billing type	`InvoicesListPage.tsx`
25	CEO approval gate: Send and Print disabled until approved	`invoice type, InvoicesListPage.tsx`
26	Receiving photo required for Delivered	`invoice record`
27	Payment evidence required to close; auto-schedule next AMC invoice	`invoice record, contracts`

Step 1.5 ID format

No.	Change	Where
28	Company-prefixed IKPM IDs: IKPM-[Abbr]-[ClientSeq]-PROP / INV / JBR / RPT-001. Needs item 47 first	`formatters.ts, pdfGenerator.ts, list pages`

PHASE 2 — New features for existing modules — 5 changes

Step 2.1 Settings and routes

No.	Change	Where
31	Settings: working hours, complaint escalation hours, service types list, random visit threshold	`SettingsPage.tsx`
32	Stub routes for HR, payroll, commission, random visits	`App.tsx`

Step 2.2 Random visits and notifications

No.	Change	Where
29	Random visit tracking: last visit per PCO, overdue flag, coordinator logs the visit	`new page, Settings`
30	All missing notification triggers (complaint, inspection, B2B requirement, random visit, overdue follow-ups)	`NotificationContext.tsx`

Step 2.3 Search

No.	Change	Where
45	**NEW** Global search for Admin and Finance: invoices, reports, challans, payments, proposals, contracts, complaints, documents, employees; results page; open the record or its folder	`lib/search.ts, SearchPage.tsx, Header.tsx`

PHASE 3 — New modules — 9 numbered, 6 to build

Step 3.1 Employee management

No.	Change	Where
42	**NEW** Employees: profile, private HR data (CNIC, bank) for HR and CEO only, documents, ID card, probation. HR or Admin enrolls PCOs here	`hr.ts, EmployeesPage.tsx, firestore.rules`
34	**REPLACED** HR module	`replaced by 42`

Step 3.2 Attendance

No.	Change	Where
43	**NEW** Daily attendance form: the coordinator marks every PCO on one screen	`AttendancePage.tsx (new)`
35	**REPLACED** Daily attendance form	`replaced by 43`

Step 3.3 Payroll

No.	Change	Where
44	**NEW** 30-day salary cycle: Finance checks, super admin approves, paid record, salary slip. Deduction rules set by the super admin in Settings	`payroll.ts, PayrollPage.tsx, SettingsPage.tsx`
36	**REPLACED** 30-day payroll summary	`replaced by 44`

Step 3.4 Document and Evidence Automation Engine

No.	Change	Where
33	Collections doc_templates, doc_procedures, doc_rules, doc_cases, doc_case_items, doc_evidence_files; pages for templates, cases, evidence	`new module`
37, 38	Not readable in the uploaded PDF (the PDF ends after item 36)	

Who does what

Role	Does
HR / Admin	Enroll PCOs and all staff as employees (CNIC, documents, ID card, salary structure, probation). Prepare payroll cycles.
Operation Coordinator	Assign PCOs to jobs. Submit reports for PCOs (photos, signed report, chemicals). Upload challan and number. Mark daily attendance. Log random visits.
Operation role	Verify reports and challans. Must not be the person who submitted that report.
Finance / Accounts	Check payroll cycles, record salary payment, invoices, payments, search.
Super admin	Approve payroll cycles, set deduction rules in Settings, approve invoices.
PCO	No login. Exists as an employee record only.

Decisions already made

- Challan means the delivery / service challan signed by the client.
- Operation and Operation Coordinator roles share the same permission (verify, upload challan). The submitter is never the verifier.
- PCOs have no login. The coordinator enters everything for them.
- The super admin decides salary deductions, set in Settings.

Still open

- Send gap list items 1 to 7, 37 and 38 (see the note on page 1).
- Folder layout: one client-first path with service type as a tag and view (recommended), or also a separate service folder.
- Finance and HR: new logins with the new Accounts and HR roles, or Admin users do this work.
- Performance score formula per role (SRS open item). The super admin can set it, but it is needed before payroll goes live.

What I found in your code first

- **Verified reports are already filed.** handleVerifyQC in JobsListPage.tsx generates the PDF and files it under the client's Service Reports/YYYY/MM/JobID folder. But if filing fails it only does console.warn while the screen still says "Report auto-filed". That is the real gap behind your request (item 41).
- **Nothing exists yet** for challan, attendance, salary or employee records. The only "challan" in the code is fbrChallanNumber on supplier tax in finance.ts, which is a different thing.
- **Search exists but is small.** The header search in Header.tsx covers customers, leads, jobs and invoices only, 4 results each, and an invoice result opens the generic invoice list, not that invoice.
- **There is no Accounts / Finance or HR role.** DEFAULT_ROLES has super_admin, admin, sales, operations, team_leader, technician. Only super_admin can edit invoices; admin has view only.
- **New modules will silently not work for built-in roles** unless they are also added to DEFAULT_ROLES in permissions.ts. setRoleDefinitions drops any module on a system role that the default list does not contain.
- **IDs use list length + 1** (generatePermanentId('CL', clients.length + 1) in dataService.ts). Two people creating at once, or one deletion, can produce duplicate IDs.
- **Items 34, 35 and 36 in the gap list overlap** with items 42, 43 and 44 here. Build these versions instead; they match your real data layer (Firestore, folders, roles).
- **There is no separate PCO portal.** A PCO today is a staff login with the technician role, who sees only their own jobs. Job reports are submitted from the report modal in JobsListPage.tsx (Team Leader and Operations can do it). Because you want PCOs enrolled only for assignment, with the coordinator doing all entry, items 48 and 49 change this model. Gap list items 11 to 14 (checklist, photos, late flag) are re-read as steps the coordinator completes for the PCO.

Confirmed decisions

- **Challan** means the delivery / service challan signed by the client for a visit, uploaded at report confirmation.
- **Operation and Operation Coordinator roles** in your code get the same permission to verify reports and upload challans. No separate Operations Manager role is needed now.
- **PCOs have no portal login.** They are enrolled as records so jobs can be assigned to them; the coordinator does everything else (items 48, 49).
- **Salary deductions** are decided by the super admin, set in Settings, and never hardcoded.

Detail pages: changes 39 to 49

Each page below gives the current behaviour, the files to change, the data fields, the rules and a Done-when check for one new change.

39 Challan number and upload at report confirmation

OPERATIONS

When the Operations Coordinator or Operations Manager confirms (verifies) a job report, they must enter the challan number and upload the challan photo or PDF. Without both, the report cannot be confirmed.

WHAT THE CODE DOES TODAY

The verify modal (showVerifyModal, verifyForm in JobsListPage.tsx) has only supervisorName, qcRating and supervisorNotes. Nothing captures a challan.

FILES TO CHANGE OR CREATE

File	What to change
`src/types/operations.ts`	Add to ServiceJob: challanNumber?, challanId?, challanFileUrl?. Add a Challan interface (below).
File	**What to change**
`src/pages/operations/JobsListPage.tsx`	Add challan number input and file upload to the verify modal. Disable the Verify button until both are present. Save through the same dataService.saveJob call, then create the challan record.
`src/features/field/SinglePhotoField.tsx`	Reuse for the challan photo. Allow PDF as well as images (check what it accepts).
`src/services/dataService.ts`	Add saveChallan(), getChallans(), findChallanByNumber(); register the new collection (see item 46).
`src/lib/permissions.ts`	Operation and Operation Coordinator roles, admin and super_admin can verify and upload challans (same permission). The CRO must not (SRS). Add the role check in the verify handler as well as hiding the button.

DATA FIELDS

Field	Type	Note
`challanNumber`	text, required	As printed on the paper. Trim spaces, keep original case for display.
`challanKey`	text, auto	Lower-case, spaces and dashes removed. Used for search and duplicate check.
`jobId, contractId`	link	From the job being verified.
`customerId, clientType, branchId`	link	Same folder fields as other records (FolderMeta).
`challanDate`	date	Date on the paper; default to service date.
`fileUrl, fileName`	text	Stored through the existing uploadFile (Cloudinary) path.
`uploadedBy, uploadedByName, uploadedAt`	auto	From the signed-in user.

RULES

- Both number and file are required to verify. No skipping.
- Duplicate check on challanKey: same number for the same client is blocked; the same number for a different client shows a warning with "Use anyway" and a reason.
- A challan is never deleted. A replacement keeps the old file and is marked superseded.
- Write an audit log line (entityType needs a new value, see item 46).

DONE WHEN

A coordinator cannot verify a job without a challan number and file; after verifying, the number is visible on the job and in the client folder.

40 Challan register and search by number

OPERATIONS / FINANCE

One list where anyone allowed (Operations, Admin, Finance) can type a challan number and land on the challan file, the job and the client.

WHAT THE CODE DOES TODAY

No register exists.

FILES TO CHANGE OR CREATE

File	What to change
`src/pages/operations/ChallansPage.tsx (new)`	Table: challan number, client, job, date, uploaded by, file link. Filters: client, date range, uploader. Search box on challan number (partial match on challanKey).
File	**What to change**
`src/App.tsx`	Route /operations/challans guarded by the jobs module.
`src/components/layout/Sidebar.tsx`	Menu entry under Field operations.
`src/pages/finance/InvoicesListPage.tsx`	Show the challan number(s) of the linked job on the invoice detail so Finance can see it without leaving the page.

RULES

- Finance gets read-only access to challans (view level on the module).
- Searching "1042" must find "CH-1042" and "ch 1042" (that is why challanKey exists).

DONE WHEN

Typing a challan number in the register or in global search (item 45) opens that challan with a link to its job and client.

41 Jobs-done report saved in the client and service folder on confirmation

DOCUMENTS

When the report is confirmed, the final report, the signed service report photo, the before and after photos and the challan are all saved in that client's folder for that job, and the person can see that it worked.

WHAT THE CODE DOES TODAY

handleVerifyQC already calls pdfGenerator.jobReportPDFBlob and fileGenerated with category service_report and subPath = job id. The failure case is hidden (console.warn) and only the PDF is filed, not the challan or the signed report photo.

FILES TO CHANGE OR CREATE

File	What to change
`src/pages/operations/JobsListPage.tsx`	In handleVerifyQC: await the filing, and if it fails show a warning toast with a Retry button and mark the job filingPending: true. Never show "auto-filed" unless it succeeded. After the PDF, also file the challan (category challan) and the signed report photo into the same job subfolder.
File	**What to change**
`src/types/folders.ts`	Add 'challan' to FileCategory.
`src/lib/folders.ts`	Add 'Challans' to CLIENT_SUBFOLDERS and map challan to it in CATEGORY_FOLDERS (dated by year). Add the label in FILE_CATEGORIES.
`src/types/document.ts`	Add challanNumber? and serviceType? to the document record so the folder view and search can show them.
`src/features/documents/DriveExplorer.tsx`	Add a "By service" view: group Service Reports by serviceType (Rodent, Termite, Fogging and so on) alongside the normal client path.
`src/lib/folderBackfill.ts, scripts/backfillFolders.ts`	Add a backfill for jobs already verified before this change.

RULES

- Path stays client-first: /Clients/Name__ID/Service Reports/YYYY/MM/JobID. Service type is a tag and a view, not a second copy of the file, so nothing is stored twice.

- Re-verifying a job files a new version and keeps the old one.
- Offline: fileGenerated already falls back to a download; keep that, but mark filingPending so the report is filed when the connection returns.

DONE WHEN

Verify a job, open the client's Service Reports folder, and find the PDF, signed report photo and challan under that job. Turn off the network, verify, and see a clear "not filed yet" status.

42 Employee management

HR (NEW)

One employee record per person (PCO, coordinator, CRO, accounts, HR), separate from the login account, with sensitive data locked to HR and the CEO. This replaces gap list item 34 for the data side.

WHAT THE CODE DOES TODAY

users/{uid} holds logins and roles only (UserProfile). Team & Access creates logins through staffService.createStaff. There is no CNIC, salary, joining date or probation tracking.

FILES TO CHANGE OR CREATE

File	What to change
`src/types/hr.ts (new)`	Employee, EmployeePrivate, EmployeePay interfaces.
File	**What to change**
`src/pages/hr/EmployeesPage.tsx (new)`	List with status filter (Hiring, Probation, Permanent, Notice, Inactive), profile page with tabs: Profile, Documents, Attendance, Salary.
`src/services/staffService.ts`	Add "Create portal login" button on an employee that calls the existing createStaff and stores the uid on the employee.
`src/lib/folders.ts`	New root /Employees/Name__EMP-ID with subfolders: CNIC & Address Proof, Agreements, Salary Slips, Other. Reuse the same safeSegment and ID_SEP helpers.
`src/lib/pdfGenerator.ts`	Add employee ID card PDF (photo, name, ID, role) and, later, offer letter.
`firestore.rules`	Three collections with different readers (see below).

DATA FIELDS

Field	Type	Note
`employees/{id}`	public to staff	Name, role, department, reportingToRole, joiningDate, status, probationEndDate, uid (login), photo.
`employee_private/{id}`	HR + super admin	CNIC, father name and CNIC, address, witnesses, bank account. Encrypt CNIC and bank fields at rest (SRS section 8).
`employee_pay/{id}`	Finance + HR + super admin	Basic salary, attendance allowance, mobile package, bike maintenance, total (auto sum).

RULES

- Never delete an employee; set status to inactive. CNIC must be unique (duplicate check).
- Splitting data across three documents is what lets Firestore rules protect CNIC and bank details while Accounts still sees salary. One document cannot do that.
- Probation end = joining date + months from Settings (default 3). Do not hardcode 3.
- Employee ID follows the IKPM format from gap list item 28 (item 47 below makes the counter safe).

DONE WHEN

HR creates an employee, a coordinator cannot read CNIC or bank data even by calling Firestore directly, and Accounts can read salary only.

43 Daily attendance form per PCO

OPERATIONS / HR

The Operations Coordinator marks each PCO every day. This data feeds payroll (item 44). Replaces gap list item 35.

WHAT THE CODE DOES TODAY

Nothing exists.

FILES TO CHANGE OR CREATE

File	What to change
`src/types/hr.ts`	Add AttendanceRecord.
File	**What to change**
`src/pages/operations/AttendancePage.tsx (new)`	One screen: pick a date, see every active PCO as a row, set Present / Late / Absent / Leave / Half day / Off, arrival time, remarks. "Mark all present" button, then fix exceptions. Save writes one record per PCO.
`src/App.tsx, Sidebar.tsx`	Route /operations/attendance under the operations menu.
`src/services/dataService.ts`	saveAttendance(), getAttendance(employeeId, from, to).
`src/pages/settings/SettingsPage.tsx`	Official working hours and the late-arrival thresholds (late minutes that count as a late mark). Deduction amounts are set in item 44, not here.

DATA FIELDS

Field	Type	Note
`id`	text	employeeId + date, e.g. EMP-0007_2026-10-30. This makes one record per person per day, so double entry is impossible.
`employeeId, date`	link, date	Date in Pakistan time.
`status`	dropdown	present, late, absent, leave, half_day, off.
`arrivalTime, lateMinutes`	time, number	lateMinutes calculated from official start time in Settings.
`remarks, markedBy, markedAt`	text, auto	Edits keep old value in the audit log.

RULES

- Coordinators can edit the last 2 days; after that only the super admin (configurable in Settings).
- PCOs have no login, so only coordinators, HR, Finance and the super admin see attendance.

DONE WHEN

Marking 20 PCOs takes one screen and one save; payroll shows the same days.

44 Salary distribution (30-day payroll)

PAYROLL (NEW)

Every 30 days the system builds a salary line per employee from attendance, approved work and deductions. Finance checks it, the CEO approves it, then Finance records the payment and files the salary slip. Replaces gap list item 36.

WHAT THE CODE DOES TODAY

Nothing exists.

FILES TO CHANGE OR CREATE

File	What to change
`src/types/hr.ts`	SalaryCycle and SalaryLine interfaces.
File	**What to change**
`src/lib/payroll.ts (new)`	Pure functions: calculateSalaryLine(employee, attendance, jobs, rules). Pure functions so they can be unit tested like the existing files in src/tests.
`src/pages/hr/PayrollPage.tsx (new)`	Create cycle (start date, 30 days), review lines, adjust with a reason, submit for approval, mark paid.
`src/lib/pdfGenerator.ts`	Salary slip PDF, filed into /Employees/Name__ID/Salary Slips via fileGenerated.

File	What to change
src/pages/approvals/ApprovalsPage.tsx	Add "salary cycle" as an approval type so the CEO approves in the existing queue.
src/pages/settings/SettingsPage.tsx	New "Payroll rules" tab visible and editable by super_admin only: deduction per late mark, per absence, per unpaid leave day (fixed amount or percent of daily wage), and how many late marks equal one absence. Payroll reads these values; nothing is hardcoded.
src/tests/payroll.test.ts (new)	Test against a hand-calculated employee, as the SRS "done when" says.

DATA FIELDS

Field	Type	Note
cycle status	dropdown	draft, verified (Finance), approved (CEO), paid.
line: base, allowances	number (PKR)	Copied from employee_pay at cycle creation, so later salary changes do not alter old cycles.
line: attendanceDays, lateMarks	number	From item 43.
line: verifiedJobs, approvedExpenses	number	Jobs with supervisorVerification.isVerified in the period.
line: performanceScore	number	From job completion and QC rating (qcRating already exists). Exact formula per role is an SRS open item.
line: deductions, incentive, net	number	Calculated from the super admin Payroll rules in Settings. Each deduction line shows which rule produced it.
paidOn, paymentMethod, paymentRef	date, text	Recording only. The SRS keeps bank transfers out of scope.

RULES

- Salary data visible to Finance, HR and super admin only.
- Once approved, a cycle is locked. Corrections go in the next cycle or as a reversal with a reason.
- Only the super admin can change deduction rules, and a rule change applies to new cycles only, never to a cycle already approved. Audit-log every change.

DONE WHEN

The cycle total matches a manual calculation for a test employee; the CEO approves; Finance marks paid; the slip appears in that employee's folder.

45 Global search for Admin and Finance

SEARCH

One search that finds invoices, reports, challans, payments, proposals, contracts, complaints, documents and employees by number, name or file name, with a results page and a link straight to the record or its folder.

WHAT THE CODE DOES TODAY

Header.tsx searches companies, individual clients, leads, jobs and invoices with simple includes() on already-loaded data, 4 hits per type, and only for modules the role can open. Invoice results link to /finance/invoices without selecting the invoice. Reports, challans, payments, proposals, contracts, complaints and documents are not searched.

FILES TO CHANGE OR CREATE

File	What to change
src/lib/search.ts (new)	Move the search out of Header.tsx into one function: search(query, session) returning typed hits. Add the missing types and match on: invoice number, receipt number, bank reference, cheque number, challan number, job ID, proposal/contract ID, complaint ID, document title and file name, client name, phone, NTN, employee name/ID (HR roles only).
File	**What to change**
src/pages/search/SearchPage.tsx (new)	Route /search?q=. Results grouped by type with filters (type, client, date range, status). Each row has "Open" and "Open folder".
src/components/layout/Header.tsx	Keep the dropdown for quick hits, but call lib/search.ts, add "See all results" to /search, and add Ctrl+K to focus. Raise hits per type for Finance and Admin.

File	What to change
`src/components/layout/Sidebar.tsx`	Add a Search entry for roles with the search module.
`src/pages/finance/InvoicesListPage.tsx`	Read ?invoice=ID (and ?job=ID like JobsListPage does) and open that invoice, so search hits land on the record.
`src/features/documents/DriveExplorer.tsx`	Accept a folder path or document id in the URL and open it.

RULES

- Search only what the person may open. Reuse can(module) exactly as Header.tsx does now; never search a collection the role cannot read.
- Admin and Finance roles need company-wide data scope or results will be missing. Check defaultScope on both roles.
- Today this runs in the browser over data already loaded, which is fine for current volume. If documents grow into the thousands, add a searchKeywords array on each record or an external index; plan for it but do not build it now.

DONE WHEN

Finance types an invoice number, challan number, bank reference or a file name and lands on the right record in one click; a coordinator cannot find salary or CNIC data.

46 Roles, modules, routes and security rules for all new modules

FOUNDATION

Do this before 39 to 45. Every new screen depends on it.

WHAT THE CODE DOES TODAY

Modules are a fixed ModuleKey list in permissions.ts, mirrored by hand in firestore.rules (defaultModulesFor) and in the dataService collection map.

FILES TO CHANGE OR CREATE

File	What to change
`src/lib/permissions.ts`	Add ModuleKey values: hr, attendance, payroll, challans, search. Add them to MODULES and to the module map for routes. Add default roles Accounts / Finance (invoices edit, tax view, payroll edit, challans view, search) and HR (hr edit, payroll view). **Add the new keys to DEFAULT_ROLES**, otherwise setRoleDefinitions strips them from system roles.
File	**What to change**
`src/types/index.ts`	Add entityType values to AuditLogEntry: challan, employee, attendance, payroll.
`src/services/dataService.ts`	Add to StateKey, COLLECTIONS, BUSINESS_KEYS, the module-to-collection read map and subscriptions: challans, employees, employee_private, employee_pay, attendance, salary_cycles.
`firestore.rules`	Add match blocks for each collection. Update defaultModulesFor with the new roles and modules. employee_private: HR and super admin only. employee_pay and salary_cycles: finance, HR, super admin. attendance: operations edit, HR view, own record for technician.
`firestore.indexes.json`	Indexes for attendance by employee and date, challans by key, salary lines by cycle.
`src/App.tsx, Sidebar.tsx`	Routes /hr/employees, /hr/payroll, /operations/attendance, /operations/challans, /search, each wrapped in guard(...).
`src/tests/firestoreRules.test.ts`	Add cases: coordinator cannot read employee_private; finance can read employee_pay; technician cannot read other attendance.
`Deploy`	npx firebase-tools deploy --only firestore (rules and indexes) after the code ships.

RULES

- Edit permissions.ts and firestore.rules together, in the same commit. If they drift, the menu shows a page the server then refuses.

DONE WHEN

Each role sees only its own new menus, and the rules tests pass.

47 Safe ID counters

FOUNDATION

Replace "collection length + 1" with a counter document updated in a Firestore transaction. Needed before employees, challans and the IKPM-prefixed document IDs (gap item 28) go live.

WHAT THE CODE DOES TODAY

generatePermanentId in formatters.ts is called with this.s.clients.length + 1 and similar in dataService.ts. Deleted or filtered records, or two users saving at the same moment, can repeat a number.

FILES TO CHANGE OR CREATE

File	What to change
`src/services/dataService.ts`	Add nextSequence(key) using runTransaction on id_counters/{key}. Use it everywhere generatePermanentId gets a length-based number.
File	**What to change**
`src/lib/formatters.ts`	Add generateDocumentId(entityType, company, clientSeq, docSeq) from gap item 28. Keep generatePermanentId for old records.
`firestore.rules`	Allow signed-in staff to update id_counters only by incrementing.

RULES

- Per-client document counters (the 001, 002 in IKPM-ABCD-001-INV-001) need one counter per client and per document type.
- Existing records keep their old IDs; do not rewrite them.

DONE WHEN

Two browsers creating a client at the same second get different IDs.

48 PCOs are enrolled records only (no PCO login)

OPERATIONS

A PCO is added once as an employee record (item 42) with role PCO and no portal login. The coordinator assigns jobs to that record. This is the "just enrolled for assignment" model.

WHAT THE CODE DOES TODAY

A PCO today must be a staff login. JobsListPage.tsx builds its assignment dropdowns from dataService.getStaff([team_leader, operations, technician]), and jobs store leadPcoId and assignedTechnicianIds as login uids. Firestore rules only let a technician see jobs listing their uid. A PCO without a login cannot be assigned at all.

FILES TO CHANGE OR CREATE

File	What to change
`src/types/operations.ts`	Add to ServiceJob: leadPcoEmployeeId?, assignedPcoEmployeeIds?. Keep leadPcoId and assignedTechnicianIds so old jobs still display.
File	**What to change**
`src/pages/operations/JobsListPage.tsx`	Replace the technician dropdowns (new job form, assign form, report modal) with active employees whose role is PCO, read from the employees collection.
`src/services/dataService.ts`	Add getEmployees({role, status}). Rewrite checkPcoConflict to compare employee IDs. As it reads now it checks assignedTechnicianNames.includes(technicianId), which compares a name list with an ID and probably never reports a clash; confirm and fix.
`src/pages/hr/EmployeesPage.tsx`	"Create portal login" stays optional and is off by default for PCOs.
`scripts/migrateTechnicians.ts (new)`	One-time: for each existing technician login, create an employee record linked by uid and fill the new job fields.
`src/pages/operations/DispatchCalendarPage.tsx`	Show PCO names from employee records.
`firestore.rules`	No change for coordinators (they already read all jobs). Keep the technician rule for anyone you later give a login.

RULES

- A PCO record needs only name, phone, joining date, status and reporting coordinator to be assignable. HR fields (CNIC, bank) can be completed later and the profile marked Incomplete, as the SRS import rules do.
- Inactive PCOs cannot be assigned but stay on old jobs and reports.

DONE WHEN

A PCO with no login appears in the assignment dropdown, is assigned to a job, and a double booking on the same date and slot is caught.

49 Coordinator submits the job report on behalf of the PCO

OPERATIONS

The coordinator completes the whole report for the PCO: pre-service checklist, before and after photos, chemicals, signed report photo, expenses and notes. The record shows who did the work and who entered it.

WHAT THE CODE DOES TODAY

The report modal (handleSubmitReport in JobsListPage.tsx) already lets staff with edit access choose a technician (reportPcoId) and stores the result on the job. It does not record who entered it, and the same person can then verify it.

FILES TO CHANGE OR CREATE

File	What to change
`src/types/operations.ts`	Add to ServiceJob: performedByPcoIds (required), submittedByUid, submittedByName, submittedAt, checklistConfirmedBy, lateSubmission, lateDays, lateReason. Add uploadedBy to PhotoEvidence.
File	**What to change**
`src/pages/operations/JobsListPage.tsx`	Report modal: required "Work done by" PCO multi-select. Pre-service checklist, before photo, signed report photo and challan barriers (gap items 11 to 13, item 39) stay, ticked and uploaded by the coordinator. Late flag (gap item 14) is calculated from the coordinator submit date and asks for a reason.
`src/pages/operations/JobsListPage.tsx (handleVerifyQC)`	Block verification when the current user is the submitter. Allow another Operation or Operation Coordinator user, admin or super admin. If a super admin verifies their own submission, allow it but log "self-verified".
`src/lib/pdfGenerator.ts`	Job report PDF: show "Work done by: PCO names" and "Entered by: coordinator name, date".
`src/pages/complaints/ComplaintsPage.tsx, src/pages/customers/*`	Trace complaints and the client "who worked here" view through performedByPcoIds, not through the submitter.
`src/lib/payroll.ts (item 44)`	Credit verified jobs and complaints to the PCO in performedByPcoIds.

RULES

- **Checks and balances:** your SRS wants one person to do the work and another to verify it. If the coordinator enters everything, the verifier must be a different person. That is why the submitter-is-not-verifier block is in this item. Confirm this is acceptable.
- Photos and the signed report arrive from the PCO (for example by WhatsApp). The coordinator uploads them, so each photo stores uploadedBy and shows upload time, not capture time. Say so in the report PDF.
- Expenses: the coordinator enters them with the receipt photo, and the senior verification step in the SRS still applies.

DONE WHEN

A coordinator enters a full report for a PCO who has no login, the PDF shows both names, and the same coordinator cannot verify it.